Samarium Cobalt Magnet Properties

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Sintered SmCo properties defined in FEMM are meant to be representative of typical examples rather than strictly based on the products from one vendor. Reasonable references are below.

SmCo magnets come in two formulations:

- *SmCo1:5* was the first developed SmCo alloy. Energy products for this alloy range from 16 to 25 MGOe. This alloy has a slightly higher electrical conductivity with 1.8 MS/m being a representative value. Based on manufacturer's specs, a relative permeability 1.03 is typical for this alloy.
- *SmCo2:17* is a more recent SmCo alloy. Energy products for this alloy are higher, ranging from 20 to 35 MGOe. Electrical conductivity is lower with 1.16 MS/m being a representative value. Based on manufacturer's specs, a relative permeability 1.1 is typical for this alloy.

Both alloys have a very high coercivity and a straight-line second-quadrant demagnetization curve.

On the basis of data from many vendors, the numerical designation of the magnet grade represents the best-case energy product in units of MGOe as measured at room temperature (20°C) for a range of energy product usually spanning a range of 2 MGOe. However, magnets are rarely delivered magnetized to this best-case magnetization. For this reason, it is assumed that the actual energy product is the numerical grade minus one in units of MGOe, i.e. a SmCo26 magnet has an energy product of 25MGOe at 20°C.

The grades assumed by FEMM are listed below in Tables 1 and 2.

Grade	BH_{max}, MGOe	B_r, T	B_r, kG	H_{cB}, kA/m	H_{cB}, kOe	σ, MS/m
SmCo18	17	0.837	8.37	647	8.13	1.8
SmCo20	19	0.885	8.85	684	8.59	1.8
SmCo22	21	0.930	9.30	719	9.03	1.8
SmCo24	23	0.973	9.73	752	9.45	1.8

Table 1: Typical SmCo1:5 Properties

Grade	BH_{max}, MGOe	B_r, T	B_r, kG	H_{cB}, kA/m	H_{cB}, kOe	σ, MS/m
SmCo24	23	1.006	10.06	728	9.15	1.16
SmCo26	25	1.049	10.49	759	9.53	1.16
SmCo28	27	1.090	10.90	789	9.91	1.16
SmCo30	29	1.130	11.30	817	10.27	1.16
SmCo32	31	1.1679	11.68	845	10.62	1.16

Table 2: Typical SmCo2:17 Properties

References

Dexter Magnetics, Samarium Cobalt Magnets, accessed 22 Aug 2019.
Bunting Magnetics Europe, Samarium Cobalt Data Sheet, accessed 18 Dec 2020.
https://en.wikipedia.org/wiki/Samarium%E2%80%93cobalt_magnet
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